

LIST OF ALL COURSES FOR INDUSTRIAL CHEMISTRY BTECH PROGRAM

The contents of the core and elective courses offered to the Industrial Chemistry BTech students can be found in this [link](#)

Course Code	Credits	Semester
CY xxxx	01-03	1-8 Sem

ENVIRONMENTAL CHEMISTRY - I

Know our environment (atmosphere composition and behavior, ecosystem, flow of energy and nutrient cycles, sustainability), Know about global warming, concise overviews of ozone depletion and atmospheric pollutants. Organic and Inorganic chemicals in environment and their impact on environment, a project on environment related topic.

Course Code	Credits	Semester
CY 1017	03	1st Sem

DYNAMICS OF CHEMICAL SYSTEMS - I

Introduction to Chemical kinetics, Basics in Electrochemistry: Electrochemical Principles and Reactions, Basic concepts of electrochemical cells and batteries, Historical background of quantum hypothesis, Wave equation, Postulates, Schrodinger equation and introduction to simple solvable problems.

Course Code	Credits	Semester
CY 1020	03	1st Sem

DYNAMICS OF CHEMICAL SYSTEMS - II

Advanced insights into chemical kinetics including Collision theory of reaction rates, Transition State Theory, Energy Storage Systems (e.g. Lead-Acid, Lithium Ion Batteries, Fuel Cells and Supercapacitors): fundamentals to applications, detailed understanding of Corrosion, electroplating, electroless plating

Course Code	Credits	Semester
CY 1021	03	2nd Sem

ENVIRONMENTAL CHEMISTRY - II

More about environment (chemistry of lithosphere, energy balance, sustainability and recycle), More on global warming (infrared absorption, molecular vibration, atmospheric window, residence time of greenhouse gases, evidences and effects of global warming), Deeper analysis of atmospheric pollution (Chemistry of CO, NOx, VOCs, SO2, Industrial smog, photochemical smog), Ozone depletion (production, catalytic destruction), Fate of organic/inorganic chemicals in natural and engineered systems (fate of polymers after use, detergents, synthetic surfactants insecticides, pesticides etc. after use), impact on physical-chemical properties of environmentally relevant compounds, Aspects of transformations in atmosphere (microbial degradation of organics- mechanism of action of DDT or analogues, environmental degradation of polymers, atmospheric lifetime, toxicity), Future challenges (CO2 sequestering, Nuclear energy), a project on environment related topic

Course Code	Credits	Semester
CY 1030	03	2nd Sem

Introduction to Quantum Chemistry

The failures of classical mechanics, Wave-particle duality, Quantization of energy, Uncertainty principle, Postulates of quantum mechanics, Free particle, Tunneling, Particle in a box, Harmonic oscillator, particle on a ring, particle on a sphere, Rigid rotor, Angular momentum, Hydrogen atom, Spin, Many electron system, term symbol, Born-Oppenheimer approximation, Approximation methods, Molecular orbital theory

Course Code	Credits	Semester
IC 1050	02	2nd Sem

Machine Learning in Chemistry

Introduction to artificial intelligence and machine learning, Dataset, Data splitting, Supervised and unsupervised machine learning, Linear and logistic regression, Kernel ridge regression, Support vector machine, K-nearest neighbor, Decision tree, random forest, other ensemble methods, neuarl network, deep learning, convolutional neural network, recurrent neural network, clustering algorithms, dimensionality reduction techniques, Chemical/molecular descriptors, Model building for problems related to Chemistry

Course Code	Credits	Semester
IC 3023	03	5th Sem

Numerical and Computational Chemistry

Introduction to python programming, Plotting, Use of modules, Solving linear equations, Root finding methods, Interpolation, Curve fitting, Numerical integration, Solving differential equations, Eigenvalue and eigenvector, Solving Newton's equation of motion, Solving 1D Schrödinger equation, Electronic structure calculations for small molecules, Calculating molecular properties, Perodic systems, Calculating properties of materials.

Course Code	Credits	Semester
IC 3013	03	5th Sem

LABORATORY EXPERIMENTS

Basic chemistry related experiments: synthesis and characterization of organic molecules such as paracetamol, aspirin and acetanilide; Chemical analysis of organic compound; quantitative experiments on the estimation of phenol, sodium in cement, percentage of iron content in haematite; Determination of Hardness of Water; Synthesis of potash alum from scrap aluminium a recycling of aluminum waste, experiments on reaction kinetics; Phase Rule; Acid Strength in Citrus Fruit; Formation Constant of KI3

Course Code	Credits	Semester
CY 1031	02	1st Sem

OFFICE ADDRESS

Department of Chemistry, IIT Hyderabad, Kandi, Sangareddy-502285, Telangana, India.

OFFICE PHONE

+91 (40) 2301 6015

OFFICE EMAIL

office@chy.iith.ac.in

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